In the Greek myth that so fascinated many of the earliest pioneers of flight, Daedalus' son Icarus died after flying too close to the sun, which melted the wax on his feathered wings. In 2001, ironically inverting the mythical experience, a NASA *Helios* ultralight flying wing powered by the sun's rays flew to the outer edge of the earth's atmosphere.

Surely one of the most extraordinary aircraft yet built, *Helios* was "piloted" by a controller on the ground and traveled at a sedate 20mph (32kph). Its wing, measuring 247ft (75.25m) and thus longer than that of a Boeing 747, was covered in solar panels that generated the electricity to drive its 14 motors. Storing electricity in fuel cells during the day allowed it to continue to operate through the night. Totally ecologically friendly, *Helios* was destined for sustained flight at the edge of space. On August 13, 2001 it set an altitude record for a propeller-driven aircraft, rising to 96,863ft (29,511m). The earth's atmosphere at that altitude is similar to the atmosphere of Mars, so the flight allowed NASA scientists to learn about the feasibility of a flying machine that might cruise the skies of the "red planet." Helios also had the potential to serve many of the functions of a satellite – in communications or weather observation, for example – at a fraction of the cost. With no need to refuel, NASA believed *Helios* would eventually be able to fly for months at a time – in effect until its parts wore out.

Helios 's brief experimental career came to an end in June 2003, when it went out of control flying over the Pacific and was lost. Yet it remained an indicator of some of the key directions that flight was taking in the 21st century. The development of unmanned aerial vehicles (UAVs) did not quite threaten to make pilots obsolete, but it was set to revolutionize many areas of aviation. And the idea of using solar power was the kind of solution designers might be forced to adopt if the pollution produced by conventionally powered aircraft in the end proved politically and socially unacceptable.

Distance traveled

Helios was above all a superb example of the constant power of aviation to amaze with unexpected feats of technological innovation, revealed time and again through the 20th century. Looking back at the distance flight advanced in its first 100 years it offered a vertiginous perspective. Any measure of aircraft performance revealed dizzying progress—speed, for example, accelerating from Glenn Curtiss's record-breaking 47mph (75kph) in 1909 to top speeds passing 400mph (640kph) in the 1930s; the breaking of the the sound barrier by the end of the 1940s; aircraft reaching Mach 2 and Mach 3 in the 1960s; the X-15 reaching Mach 6.7 in 1967; and X-43A scramjet hitting Mach 9.6 in 2004.

Similar progress could be traced in range and payload. The propeller-driven airliners of the 1930s and 1940s could fly 600–1,800 miles (1,000–3,000km); jet airliners of the late 1950s had a maximum range of around 3,720 miles (6,000km); in the early 1970s the Boeing 747 raised this figure to over 6,210 miles (10,000km); by the early 21st century, the latest passenger aircraft could fly 8,700–11,200 miles (14,000–18,000 km) non-stop. The Lockheed C-5 transport introduced at the end of the 1960s could carry about 100 times the payload of a World War I bomber, and has itself been far surpassed by transports such as the extraordinary Airbus Beluga series, capable of carrying cargoes well in excess of 50 ton.

GENTLE GIANT
NASA's Helios *remote-controlled ultralight flying wing makes its gentle progress at around 20 mph (32 kph). Powered by sunlight,* Helios *was one of a new breed of environmentally friendly aircraft. It was designed for sustained flight at the limits of the earth's atmosphere.*